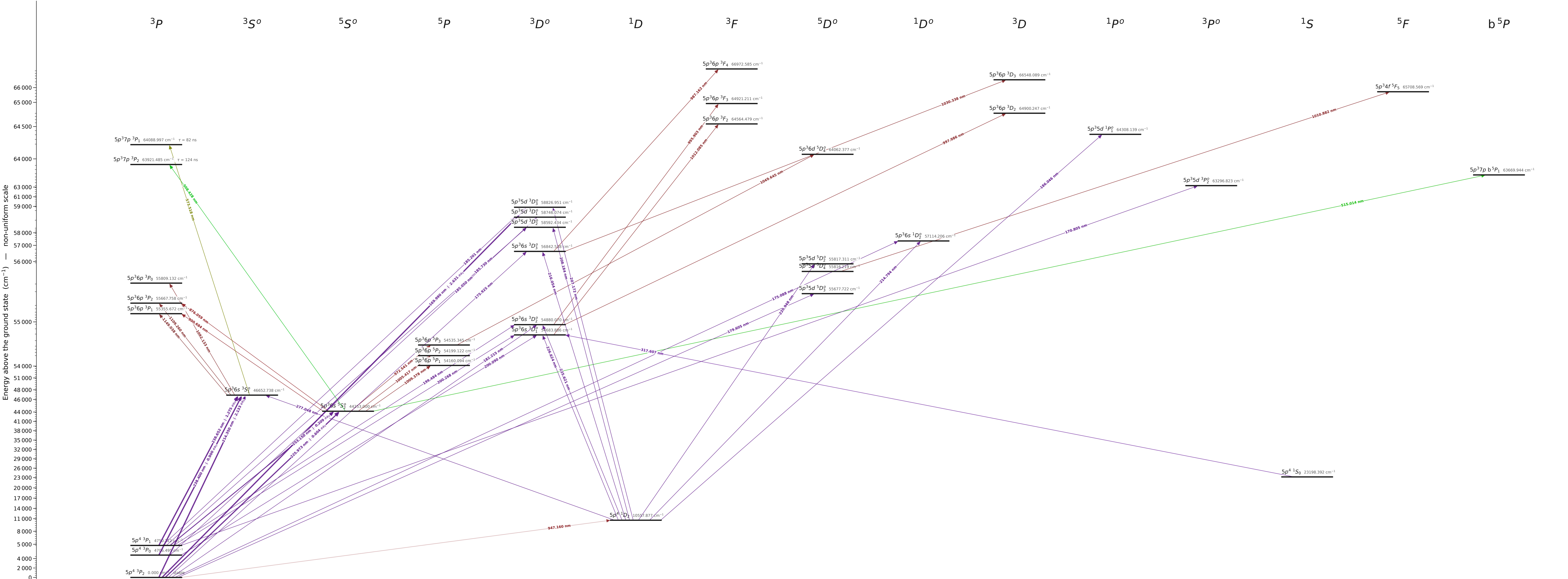
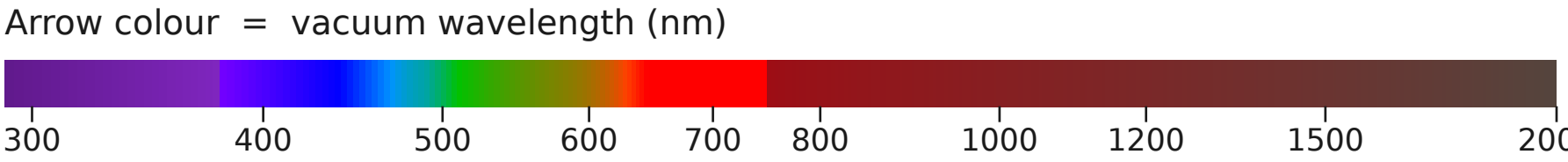


Tellurium

Te (Te I). Strongest classified transitions below 2 μm (NIST): 35 levels, 46 lines.

Te⁺ ionisation limit 72669.01 cm⁻¹ = 9.00981 eV (137.610 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J \| e r \| J' \rangle |$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J \ e r \ J' \rangle $ (eao)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
5p ⁴ 3P ₂ - 5p ³ 6s 3S ₀ ^o	214.3497	214.2822	1398.6139	2.133	3.120e+08	-	-	first step of the ISOLDE RILIS resonance-ionization schem...
5p ³ 6s 3S ₀ ^o - 5p ³ 7p 3P ₁	573.5175	573.3585	522.7259	-	-	1.941	-	second step of the ISOLDE RILIS resonance-ionization sche...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).